

Yuni Wu

South San Francisco, CA

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Summary

Machine learning engineer with experience building classification systems, evaluation pipelines, and reliability-focused ML workflows across healthcare, robotics, and cloud-native AI applications.

Technical Skills

Languages: Python, Go, SQL, R

Machine Learning: CNNs (ResNet), Transformers, hierarchical classification, class imbalance handling, targeted oversampling

Evaluation: cross-validation, ROC-AUC, F1-score, calibration, ablation workflows, benchmarking, offline evaluation

Diagnostics: inference latency, residual diagnostics, model reliability, scenario-level analysis, deployment tradeoff evaluation

Systems: Cloud Run inference evaluation, Docker, Pub/Sub, Cloud Storage, BigQuery, FastAPI, structured logging

Projects

VLM-Based Social Navigation for Quadruped

[project]

- Developed a hybrid robot navigation system combining behavior cloning with VLM-based decision rules.
- Achieved 0.11 MAE and $R^2 = 0.62$ for vision-based policy prediction under trajectory-split offline evaluation.
- Modeled human interaction context using Qwen/InternVL to generate STOP/FORWARD/LEFT/RIGHT/REVIEW actions.
- Developed scenario-level diagnostics and benchmark workflows for VLM-based navigation reliability analysis.
- Identified rotational instability ($P99 \approx 0.80$) through residual diagnostics and deployment-oriented scenario analysis.

NailAI — Serverless AI Microservice Platform

[demo]

- Built event-driven AI inference workflows using Cloud Run, Pub/Sub, and Cloud Storage for asynchronous image inference.
- Achieved 0% failure at 200 concurrent users with idempotent orchestration, retry handling, and Pub/Sub job routing.
- Implemented monitoring, regression validation, and structured logging workflows for Cloud Run inference evaluation.

EmotionFlow — LLM-Based Recommendation System

[demo]

- Built LLM evaluation and prompt optimization workflows for real-time recommendation and response quality analysis.
- Reduced hallucination by 34% and increased recommendation acceptance to 57% through feedback-driven evaluation loops.

Experience

Machine Learning Engineer Intern

May 2025 – Aug 2025

MyEdMaster LLC

- Built a 22-class medical image classification workflow on 50K+ samples, achieving 0.92 F1-score through repeatable model evaluation.
- Improved minority-class F1 by +30% using hierarchical classification, targeted oversampling, and class-imbalance evaluation.
- Evaluated ResNet deployment tradeoffs through ablation workflows, reducing inference latency by 5× from 0.15s to 0.03s per image.
- Designed evaluation pipelines with cross-validation, ROC-AUC, and calibration to measure model reliability under class imbalance.

Graduate Research Assistant

Sept 2025 – May 2026

CAIRO Lab, University of Colorado Boulder

- Built ROS/rosbag telemetry pipelines to transform raw robot data into structured state-action datasets for offline evaluation.
- Processed ROS telemetry and perception outputs into state-action datasets for navigation analysis, reliability evaluation, and inference latency benchmarking.

Software Engineer

June 2021 – Aug 2022

Shopee Co. Ltd.

- Engineered validation and anomaly detection systems across 800+ production scenarios, reducing manual QA workload by 80%.
- Built data quality monitoring pipelines to detect regression anomalies and surface near real-time reporting inconsistencies, accelerating issue detection by 78%.

Education

University of Colorado Boulder, MS, Data Science GPA: 3.76/4.0

May 2026

Soochow University, BS, Data Science GPA: 3.7/4.0

June 2021